

RAJVANSH MALHOTRA

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Education

National Institute of Technology, Kurukshetra
B.Tech, Computer Engineering

Aug 2023 – Jun 2027

Professional Experience

Samvedya AI, in collaboration with AIIMS Delhi
AI/ML Intern

Mar 2025 – May 2025

- Built an LLM pipeline converting PNG-based digital designs into SVG format, integrating image analysis, LLM orchestration, and API pipelines to automate a visual abstraction workflow.
- Delivered a functional MVP within 3 weeks using Python and modern ML techniques, streamlining visual abstraction for clinical research.
- Engineered a RAG pipeline for context-aware answers in a chat-based system, achieving 80% accuracy on clinically relevant visual abstracts at AIIMS Delhi.

National Institute of Technology, Kurukshetra
Machine Learning Research Intern

Jun 2024 – Dec 2024

- Extracted image features with VGG19 in TensorFlow and selected the top 1,000 impactful features via a Random Forest classifier.
- Improved model accuracy from 60% to 82% by training a Vision Transformer on the dataset.
- Optimized predictions by integrating Golden Ratio and eye-spacing calculations via OpenCV.
- Reached state-of-the-art accuracy with a custom ANN on structured data, reducing training time and inference cost.

Projects

BLR Parking Intelligence

Flipkart Gridlock Hackathon | July 2026

- Built an end-to-end parking enforcement AI over 298,443 GPS violation records, ranking hotspot clusters by urgency and predicting congestion propagation corridors through a hybrid RAG pipeline backed by Qwen3-8B streaming generation.
- Redesigned the LightGBM LambdaRank label to remove bias, replacing arbitrary heuristic scores and patrol-history features with growth-rate quartiles grounded in Daganzo's LWR kinematic-wave theory.
- Encoded cluster instability as a Hungarian-matched merge/diverge flag, improving hotspot ranking stability across time windows.
- Assembled a hybrid RAG corpus across 5 document types, fusing dense embeddings with TF-IDF and hard-injecting time-bucket summaries to prevent vocabulary-mismatch misses on decision-critical documents.
- Parallelized step-back reasoning with retrieval to cut time-to-first-token, then streamed grounded answers with an OSRM 5-stop patrol route appended.
- Ran a 3-model LLM fact-checking jury asynchronously after the stream closed, adding verification with zero added perceived latency; verdicts fed a GitHub-persisted RLHF loop that distilled repeated rejections into correction rules for future prompts.

Yojana AI

July 2026

- Built a bilingual (Hindi/English) voice RAG chatbot for government welfare scheme discovery, cutting end-to-end pipeline latency from 33s to under 5s.
- Diagnosed three serial LLM bottlenecks across preprocessing, routing, and generation, then resolved them by routing auxiliary calls to Groq llama-3.1-8b-instant (~800 tok/s vs ~50 tok/s), reserving the 72B model for multi-constraint queries.
- Collapsed 4 serial LLM preprocessing calls into a single structured JSON prompt, then parallelized retrieval, classification, and Firecrawl search in a ThreadPoolExecutor, reaching a 2.82s combined wall time.
- Replaced buffer-then-synthesize TTS with server-side sentence-boundary detection, firing TTS per sentence as tokens arrived to cut perceived audio latency from 20s to ~2s.
- Dropped a URL-scrape step that timed out on every request in favor of Firecrawl's native search snippets, removing a silent 4-second penalty per query.

- Resolved garbled Hindi TTS by separating romanization from translation, switching to English generation with Gemini-2.5-flash translation at the TTS layer, preserving scheme names verbatim.

Awaaz

Rumik AI Hackathon, Top 50 / May 2026

- Reduced perceived voice AI latency from 46s to under 8s (first audio chunk) by profiling and eliminating bottlenecks across 12 targeted optimizations spanning async parallelism, connection reuse, model swaps, and streaming redesign.
- Decoupled LLM inference from speaker identification using `asyncio.gather`, cutting $\sim 500\text{ms}$ from the critical path by running recognition concurrently with generation.
- Replaced per-request TTS client creation with a persistent `httpx.AsyncClient`, eliminating repeated TLS handshake overhead ($\sim 400\text{ms}/\text{request}$).
- Rearchitected the pipeline around persistent WebSocket streaming, enabling token streaming from LLM through live TTS synthesis to immediate browser playback.
- Swapped Groq Whisper for Sarvam Saaras v3 Codemix STT, improving Hinglish transcription accuracy and removing retry overhead.
- Redesigned the model's role from full conversational generation to lightweight emotion tagging, cutting output tokens 60% ($512 \rightarrow 200$) and accelerating downstream TTS.
- Built a personality-driven voice personalization system, encoding user traits into onboarding profiles that dynamically generate LLM system prompts and TTS voice descriptions.

Battery Health Optimization

Research Work, IEEE / Mar 2026

- Deployed a Physics-Informed Residual Neural Network (RPNN) on an ESP32 microcontroller for real-time SOC/SOH estimation, embedding Coulomb-counting differential equations directly into the training loss.
- Trained with a three-component loss combining data-driven MSE, ODE physics residual via automatic differentiation, and boundary constraints enforcing SOC within $[0, 100]\%$, optimized with AdamW at physics-loss weight $\lambda = 0.1$.
- Compressed the PyTorch RPNN into a C++ TinyML engine for edge deployment, removing cloud dependency for safety-critical battery diagnostics.

Gotham Orbital

Agentic Satellite Intelligence Platform / Feb 2026

- Engineered `optimum_ReAct`, a custom agentic AI library from scratch, deploying 3 concurrent LLM agents (ATLAS, ORBITAL-1, ANALYST-1) with isolated context windows and role-specific system prompts, inspired by Palantir's Gotham.
- Designed HybridMemory, combining BM25 retrieval, sentence-transformer semantic search, and a weighted knowledge graph with multi-factor eviction (importance, recency decay, access frequency, connection weight).
- Deployed the production backend on AWS EC2 with Docker containerization, `asyncio` Semaphore concurrency controls, and exponential backoff retry (15/30/60s) for Groq API fault tolerance.
- Shipped an end-to-end MLOps pipeline via GitHub Actions CI/CD.

Rizzler

LLM Fine-Tuning Project / Jun 2025

- Fine-tuned Meta LLaMA-3 8B using QLoRA via Unsloth for efficient low-resource conversational training.
- Applied LoRA adapters to key transformer components (`q_proj`, `v_proj`) with $r = 64$ and 4-bit quantization to reduce memory usage while preserving performance.
- Tuned generation quality through temperature and `top_p` sampling sweeps.

Technical Skills

Languages: Python, C/C++

ML / AI: Machine Learning, Deep Learning, NLP, LLMs, LLM Fine-tuning, RAG, Prompt Engineering, Scikit-learn, TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV

Tools & Platforms: AWS EC2, Docker, GitHub Actions, MLflow, ETL, SQL, Claude Code

Coursework: Data Structures & Algorithms, OS, DBMS, Computer Networks, Computer Architecture, Software Engineering, Artificial Intelligence

Achievements

- Amazon ML Challenge: Directed a team to the top 97.2 percentile, reducing SMAPE from 100 to 48 within 2 days.
- Solved 300+ problems on LeetCode and other competitive programming platforms.
- Taekwondo: 1 silver medal, 3 gold medals.